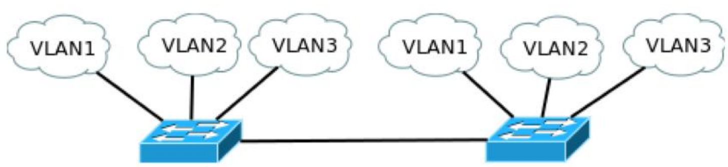
Guia 3– VLANs

Objectivos:
- Definição de VLANs
- Routing entre VLANs
- Utilização de switches L2 e L3
- Desenho e interligação de redes de Acesso e Distribuição
- Estabelecimento e configuração de trunks

1 - Definição de VLANs

1. **(LEIA O EXERCÍCIO TODO PRIMEIRO ANTES DE COMEÇAR A RESOLVÊ-LO!!!)**
Usando o GNS3, monte a rede da figura. Em vez das núvens pode utilizar VPCs ou outra entidade qualquer que simule um host (aconselha-se, por exemplo, um router c3725 configurado com um endereço IP na interface).
Configure três VLANs nos switches:
- Ports 1-2: VLAN1 (sub-rede 10.1.1.0/24)
- Ports 3-4: VLAN2 (sub-rede 10.2.2.0/24)
- Ports 5-6: VLAN3 (sub-rede 10.3.3.0/24)
- Ports 7-8: Inter-switch/Tagged/802.1Q (native VLAN 1)



NOTA:
Para replicar um switch Layer2 no GNS3 aconselha-se antes a utilização de um router c3725 configurado com um modulo de switching (NM- 16SW). Esse modulo adiciona interfaces de switching ao router, que têm o protocolo IP desligado (portos f1/0 a f1/15). Esta combinação é conhecida por “EtherSwitch router”.

Nota: O switch básico do GNS3 “Ethernet Switch” não supporta o Spanning Tree Protocol.

Configurar um “Ethernet Switch” do GNS3: Usar a interface gráfica.
Configurar um “EtherSwitch router” como sendo um switch L2:

```
EtherSwitch# vlan database
EtherSwitch(vlan)# vlan 1
EtherSwitch(vlan)# vlan 2
EtherSwitch(vlan)# vlan 3
EtherSwitch(vlan)# exit
EtherSwitch# configure terminal
EtherSwitch(config)# no ip routing
EtherSwitch(config)# interface f1/1
EtherSwitch(config-if)# switchport mode access
```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```
EtherSwitch(config-if)# switchport access vlan 1
EtherSwitch(config-if)# interface f1/2
EtherSwitch(config-if)# switchport mode access
EtherSwitch(config-if)# switchport access vlan 1
EtherSwitch(config-if)# interface range fastEthernet 1/3 - 4
EtherSwitch(config-if-range)# switchport mode access
EtherSwitch(config-if-range)# switchport access vlan 2
EtherSwitch(config-if-range)# interface range fastEthernet 1/5 - 6
EtherSwitch(config-if-range)# switchport mode access
EtherSwitch(config-if-range)# switchport access vlan 3
EtherSwitch(config-if-range)# interface range fastEthernet 1/7 - 8
EtherSwitch(config-if-range)# switchport mode trunk        !Defines   as   Trunk
port
EtherSwitch(config-if-range)# switchport trunk encapsulation dot1q      !By
default all
```

Note: Comando a usar para mostrar as VLANs existentes: `show vlan-switch`

Soluções para problemas:

Nota: Se aparecer "(no device available)" tente adicionar pelo menos mais 1MB ao PCMCIA slot 0, na configuração do router (configuração gráfica do router no GNS3).

Troubleshooting 2: Verifique se todas as interfaces com ligações estão ativas, com o comando:
`show ip interface brief`
Coloque terminais (VPCs, hosts, núvens, etc.) nas diferentes VLANs e teste a conetividade.

2. Grave as configurações dos equipamentos, bem como o projecto.

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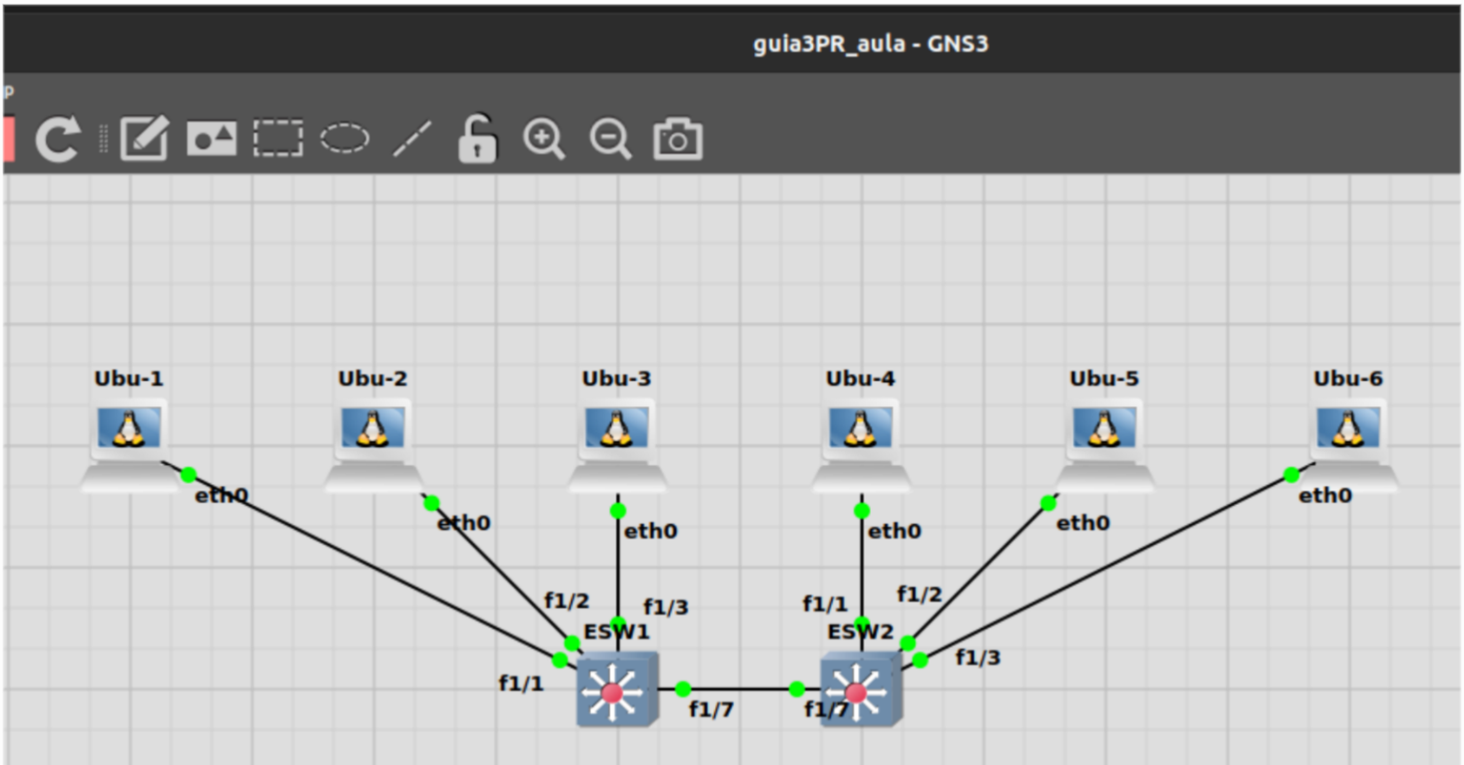
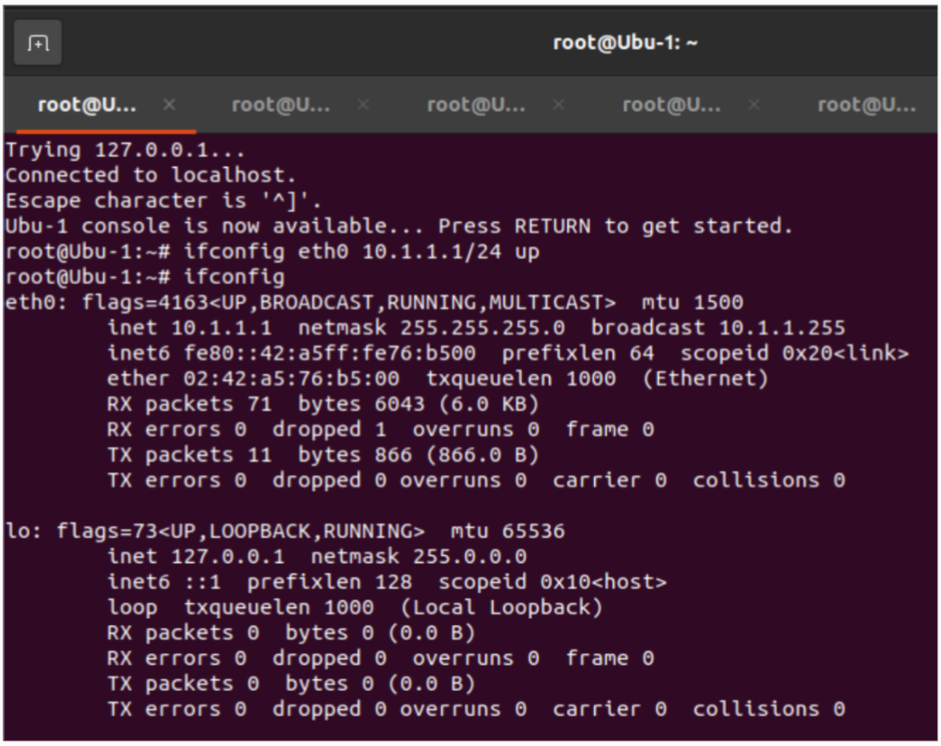
Definição de VLANs

1.





```
root@Ubu-4: ~
root@U...  root@U...  root@U...  root@U...  root@U...
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
Ubu-4 console is now available... Press RETURN to get started.
root@Ubu-4:~# ifconfig eth0 10.1.1.2/24 up
root@Ubu-4:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.1.1.2  netmask 255.255.255.0  broadcast 10.1.1.255
        inet6 fe80::42:7ff:fe6c:7300  prefixlen 64  scopeid 0x20<link>
        ether 02:42:07:6c:73:00  txqueuelen 1000  (Ethernet)
        RX packets 253  bytes 17629 (17.6 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 12  bytes 936 (936.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
root@Ubu-2: ~
root@U...  root@U...  root@U...  root@U...  root@U...
rying 127.0.0.1...
onnected to localhost.
scape character is '^]'.
bu-2 console is now available... Press RETURN to get started.
oot@Ubu-2:~# ifconfig eth0 10.2.2.1/24 up
oot@Ubu-2:~# ifconfig
th0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.2.2.1  netmask 255.255.255.0  broadcast 10.2.2.255
        inet6 fe80::42:27ff:fe3e:c300  prefixlen 64  scopeid 0x20<link>
        ether 02:42:27:3e:c3:00  txqueuelen 1000  (Ethernet)
        RX packets 81  bytes 5648 (5.6 KB)
        RX errors 0  dropped 1  overruns 0  frame 0
        TX packets 10  bytes 796 (796.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
root@Ubu-5: ~
root@U...  root@U...  root@U...  root@U...  root@U...
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
Ubu-5 console is now available... Press RETURN to get started.
root@Ubu-5:~# ifconfig eth0 10.2.2.2/24 up
root@Ubu-5:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.2.2.2  netmask 255.255.255.0  broadcast 10.2.2.255
        inet6 fe80::42:87ff:feea:3a00  prefixlen 64  scopeid 0x20<link>
        ether 02:42:87:ea:3a:00  txqueuelen 1000  (Ethernet)
        RX packets 71  bytes 5028 (5.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 10  bytes 796 (796.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
root@Ubu-3: ~
root@U...  root@U...  root@U...  root@U...  root@U...
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
Ubu-3 console is now available... Press RETURN to get started.
root@Ubu-3:~# ifconfig eth0 10.3.3.1/24 up
root@Ubu-3:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.3.3.1  netmask 255.255.255.0  broadcast 10.3.3.255
        inet6 fe80::42:90ff:fe20:c000  prefixlen 64  scopeid 0x20<link>
        ether 02:42:90:20:c0:00  txqueuelen 1000  (Ethernet)
        RX packets 31  bytes 2171 (2.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 9  bytes 726 (726.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
Ubu-6 console is now available... Press RETURN to get started.
root@Ubu-6:~# ifconfig eth0 10.3.3.2/24 up
root@Ubu-6:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.3.3.2  netmask 255.255.255.0  broadcast 10.3.3.255
        inet6 fe80::42:aaff:feb6:d400  prefixlen 64  scopeid 0x20<link>
        ether 02:42:aa:b6:d4:00  txqueuelen 1000  (Ethernet)
        RX packets 26  bytes 1861 (1.8 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 9  bytes 726 (726.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

OU, se quisermos que a configuração fique guardada (em cada PC Linux Docker), fazer:

cat /etc/network/interfaces
nano /etc/network/interfaces
descomentar a configuração que queremos (neste caso a estática), e atribuir IP e máscara.

reload no GNS3, graficamente, e assim guarda a configuração do IP

```
GNU nano 7.2                    /etc/network/interfaces *
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.1.1.1
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-1
```

```
root@Ubu-1:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.1.1.1
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-1
```

```
root@Ubu-4: ~
root@Ubu-1: ~                    root@Ubu-4: ~
 GNU nano 7.2                    /etc/network/interfaces *
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.1.1.2
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-4
```

```
root@Ubu-4:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.1.1.2
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-4
```

root@Ubu-2: ~

root@Ubu-1: ~ | root@Ubu-4: ~ | root@Ubu-2: ~ | root@Ubu

```
  GNU nano 7.2                     /etc/network/interfaces *
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.2.2.1
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-2
```

```
root@Ubu-2:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.2.2.1
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-2
```

root@Ubu-5: ~

root@Ubu-1: ~ | root@Ubu-4: ~ | root@Ubu-2: ~ | root@Ubu

```
  GNU nano 7.2                     /etc/network/interfaces *
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.2.2.2
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-5
```

```
root@Ubu-5:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.2.2.2
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-5
```

```
root@Ubu-3: ~
root@U...   root@U...   root@U...   root@U...   root@U...   root
  GNU nano 7.2                    /etc/network/interfaces *
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.3.3.1
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-3
```

```
root@Ubu-3:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.3.3.1
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-3
```

root@Ubu-6: ~

root@U... × root@U... × root@U... × root@U... × root@U... × root@U

```
  GNU nano 7.2                          /etc/network/interfaces *
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.3.3.2
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-6
```

```
root@Ubu-6:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
        address 10.3.3.2
        netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname Ubu-6
```

ESW1:

```
ESW1#vlan database
ESW1(vlan)#vlan 1
VLAN 1 modified:
ESW1(vlan)#vlan 2
VLAN 2 added:
    Name: VLAN0002
ESW1(vlan)#vlan 3
VLAN 3 added:
    Name: VLAN0003
ESW1(vlan)#exit
APPLY completed.
Exiting....
```

```
ESW1(config)#int f1/1
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 1
ESW1(config-if)#int f1/2
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 2
ESW1(config-if)#int f1/3
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 3
ESW1(config-if)#int f1/7
ESW1(config-if)#switchport mode trunk
ESW1(config-if)#switchport trunk encapsulation dot1q
ESW1(config-if)#
ESW1(config-if)#
ESW1(config-if)#exit
ESW1(config)#exit
ESW1#
*Mar  1 00:27:38.863: %SYS-5-CONFIG_I: Configured from console by console
ESW1#show vlan-switch

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa1/0, Fa1/1, Fa1/4, Fa1/5
                                                Fa1/6, Fa1/7, Fa1/8, Fa1/9
                                                Fa1/10, Fa1/11, Fa1/12, Fa1/13
                                                Fa1/14, Fa1/15
2    VLAN0002                         active    Fa1/2
3    VLAN0003                         active    Fa1/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        1002   1003
2    enet  100002     1500  -      -      -        -    -        0      0
3    enet  100003     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        1      1003
1003 tr    101003     1500  1005   0      -        -    srb      1      1002
1004 fdnet 101004     1500  -      -      1        ibm  -        0      0
1005 trnet 101005     1500  -      -      1        ibm  -        0      0
```

```
ESW1(config)#int f1/7
ESW1(config-if)#shut
ESW1(config-if)#no  s
*Mar  1 00:26:05.487: %DTP-5-NONTRUNKPORTON: Port Fa1/7 has become non-trunk
ESW1(config-if)#no  shuut
*Mar  1 00:26:06.983: %LINK-5-CHANGED: Interface FastEthernet1/7, changed state to administrativ
ely down
*Mar  1 00:26:07.983: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/7, changed s
tate to down
ESW1(config-if)#no  shut
ESW1(config-if)#no  shutdown
```

```
ESW1#wr
Building configuration...
```

ESW2:

```
ESW2#vlan database
ESW2(vlan)#vlan 1
VLAN 1 modified:
ESW2(vlan)#vlan 2
VLAN 2 added:
    Name: VLAN0002
ESW2(vlan)#vlan 3
VLAN 3 added:
    Name: VLAN0003
ESW2(vlan)#exit
APPLY completed.
Exiting....
```

```
ESW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ESW2(config)#no ip routing
ESW2(config)#int f1/1
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 1
ESW2(config-if)#int f1/2
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 2
ESW2(config-if)#int f1/3
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 2
ESW2(config-if)#no switchport access vlan 2
ESW2(config-if)#no switchport access vlan 3
ESW2(config-if)#int f1/7
ESW2(config-if)#switchport mode trunk
ESW2(config-if)#switchport trunk encapsulation dot1q
ESW2(config-if)#exit
ESW2(config)#exit
ESW2#
*Mar  1 00:33:16.307: %SYS-5-CONFIG_I: Configured from console by console
ESW2#show vlan-switch

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa1/0, Fa1/1, Fa1/3, Fa1/4
                                                Fa1/5, Fa1/6, Fa1/7, Fa1/8
                                                Fa1/9, Fa1/10, Fa1/11, Fa1/12
                                                Fa1/13, Fa1/14, Fa1/15
2    VLAN0002                         active    Fa1/2
3    VLAN0003                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        1002   1003
2    enet  100002     1500  -      -      -        -    -        0      0
3    enet  100003     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        1      1003
1003 tr    101003     1500  1005   0      -        -    srb      1      1002
1004 fdnet 101004     1500  -      -      1        ibm  -        0      0
1005 trnet 101005     1500  -      -      1        ibm  -        0      0
```

```
ESW2(config-if)#shutdown
ESW2(config-if)#noshutdown
```

Colar texto das configurações de cada EtherSwitch (com sh run):

ESW1#sh run
Building configuration...

Current configuration : 2628 bytes
!
version 12.4

```
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
no service dhcp
!
hostname ESW1
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip routing
no ip icmp rate-limit unreachable
no ip cef
!
!
!
!
no ip domain lookup
ip auth-proxy max-nodata-conns 3
ip admission max-nodata-conns 3
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
vtp file nvram:vlan.dat

!
!
ip tcp synwait-time 5
!
!
```

```
!
!
!
interface FastEthernet0/0
 description *** Unused for Layer2 EtherSwitch ***
 no ip address
 no ip route-cache
 shutdown
 duplex auto
 speed auto
!
interface FastEthernet0/1
 description *** Unused for Layer2 EtherSwitch ***
 no ip address
 no ip route-cache
 shutdown
 duplex auto
 speed auto
!
interface FastEthernet1/0
 duplex full
 speed 100
!
interface FastEthernet1/1
 duplex full
 speed 100
!
interface FastEthernet1/2
 switchport access vlan 2
 duplex full
 speed 100
!
interface FastEthernet1/3
 switchport access vlan 3
 duplex full
 speed 100
!
interface FastEthernet1/4
 duplex full
 speed 100
!
interface FastEthernet1/5
 duplex full
 speed 100
!
```

```
interface FastEthernet1/6
 duplex full
 speed 100
!
interface FastEthernet1/7
 switchport mode trunk
 duplex full
 speed 100
!
interface FastEthernet1/8
 duplex full
 speed 100
!
interface FastEthernet1/9
 duplex full
 speed 100
!
interface FastEthernet1/10
 duplex full
 speed 100
!
interface FastEthernet1/11
 duplex full
 speed 100
!
interface FastEthernet1/12
 duplex full
 speed 100
!
interface FastEthernet1/13
 duplex full
 speed 100
!
interface FastEthernet1/14
 duplex full
 speed 100
!
interface FastEthernet1/15
 duplex full
 speed 100
!
interface Vlan1
 no ip address
 no ip route-cache
 shutdown
```

```
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
banner exec ^C

***************************************************************
This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!
***************************************************************

^C
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
```

```
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
end
```

```
ESW2#sh run
Building configuration...

Current configuration : 2628 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
no service dhcp
!
hostname ESW2
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip routing
no ip icmp rate-limit unreachable
no ip cef
!
!
!
!
no ip domain lookup
ip auth-proxy max-nodata-conns 3
ip admission max-nodata-conns 3
!
!
!
```

```
!
!
!
!
!
!
!
!
!
!
!
!
vtp file nvram:vlan.dat

!
!
ip tcp synwait-time 5
!
!
!
!
!
interface FastEthernet0/0
 description *** Unused for Layer2 EtherSwitch ***
 no ip address
 no ip route-cache
 shutdown
 duplex auto
 speed auto
!
interface FastEthernet0/1
 description *** Unused for Layer2 EtherSwitch ***
 no ip address
 no ip route-cache
 shutdown
 duplex auto
 speed auto
!
interface FastEthernet1/0
 duplex full
 speed 100
!
interface FastEthernet1/1
 duplex full
 speed 100
```

```
!
interface FastEthernet1/2
 switchport access vlan 2
 duplex full
 speed 100
!
interface FastEthernet1/3
 switchport access vlan 3
 duplex full
 speed 100
!
interface FastEthernet1/4
 duplex full
 speed 100
!
interface FastEthernet1/5
 duplex full
 speed 100
!
interface FastEthernet1/6
 duplex full
 speed 100
!
interface FastEthernet1/7
 switchport mode trunk
 duplex full
 speed 100
!
interface FastEthernet1/8
 duplex full
 speed 100
!
interface FastEthernet1/9
 duplex full
 speed 100
!
interface FastEthernet1/10
 duplex full
 speed 100
!
interface FastEthernet1/11
 duplex full
 speed 100
!
interface FastEthernet1/12
```

```
 duplex full
 speed 100
!
interface FastEthernet1/13
 duplex full
 speed 100
!
interface FastEthernet1/14
 duplex full
 speed 100
!
interface FastEthernet1/15
 duplex full
 speed 100
!
interface Vlan1
 no ip address
 no ip route-cache
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!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
banner exec ^C

***********************************************************************
This is a normal Router with a SW module inside (NM-16ESW)
```

```
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 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
end
```
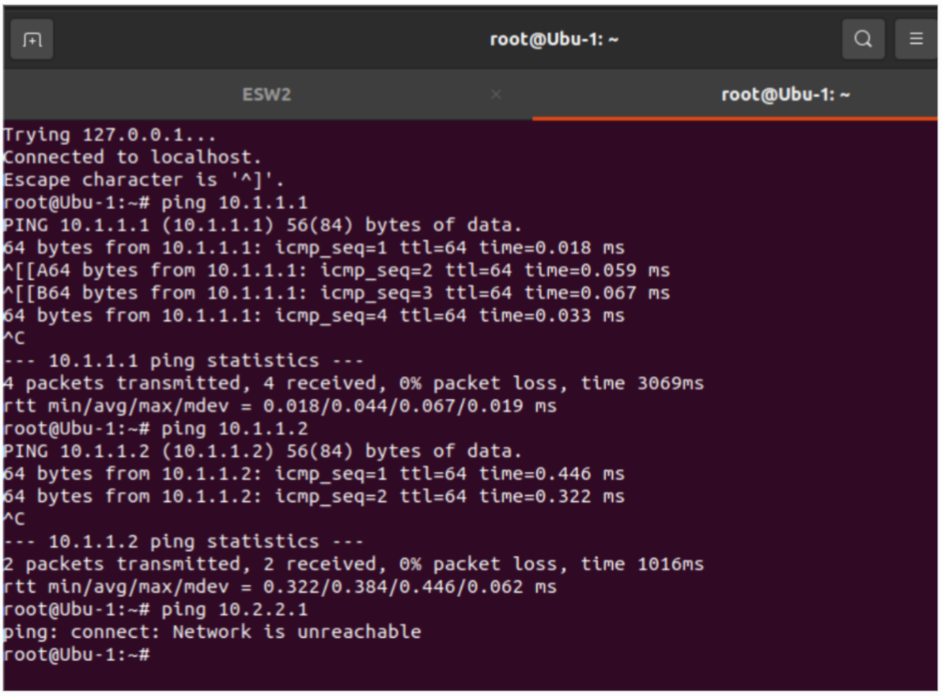

Troubleshooting 2:



2. Guardado.

Office chinês: WPS

Nos switches de L3:

Se ao guardar os projetos e abrir de novo e as vlans não funcionarem, SÓ criar a vlan database e criar os nomes das vlans de novo, e depois fica tudo a funcionar normalmente.